

New Experiment Results

Interleaved Training v3 & Sequential Dead Neuron Reinitialisation
Representation Learning in Atari RL — Progress Update

Vinol Tauro
taurov@tcd.ie

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This note reports results from two new experiments run following our meeting on 13 June. Both experiments were implemented as requested and trained for 2 M total environment steps on an NVIDIA L4 GPU.

Context: Existing Baselines

For reference, the four existing sequential conditions (all trained on Pong for 2 M steps, then Breakout for 2 M steps):

Condition	Pong (start)	Pong (end)	Dead neurons (end)
No freeze (full fine-tune)	+9.1	−20.8	87.3%
Freeze conv only	+9.1	−14.0	83.6%
Freeze conv + fc_repr (all)	+9.1	+7.2	87.5%

Table 1: Sequential transfer baselines. Pong reward measured via chimera evaluation (current backbone + original Pong head). Dead neuron threshold: >95% zero activation.

The key observation from the baselines: freezing fc_repr (the 512-dim penultimate layer) is sufficient to preserve Pong performance almost perfectly. Allowing fc_repr to update causes rapid catastrophic forgetting.

Experiment 1: Interleaved Training v3

Design

Following your suggestion, this experiment revises the previous interleaved design (v2) on two dimensions:

1. **Episode-level game switching.** In v2, the active game switched every 1 000 environment steps, cutting episodes mid-way. In v3, the game only switches at a natural episode boundary. This avoids the truncation artefact.
2. **Single joint replay buffer.** In v2, Pong and Breakout each had their own replay buffer sampled independently. In v3, all transitions from both games are stored in one shared buffer, tagged with a `game_id`. When a training batch is sampled, Pong transitions are routed through the Pong head and Breakout transitions through the Breakout head; the loss from both subsets is summed before a single backward pass. This means the shared backbone (conv + fc_repr) receives gradients from both games simultaneously.

Architecture: AtariCNNTwoHead — shared conv + fc_repr (512-dim), with separate output heads: `fc_out_pong` ($\rightarrow 6$ actions) and `fc_out_breakout` ($\rightarrow 4$ actions). Trained from scratch (no Pong checkpoint).

Budget: 1 M environment steps per game = 2 M total. Learning rate 10^{-4} , buffer capacity 100 k, batch size 32. Game selection at each episode boundary is step-balanced: always switch to whichever game has accumulated fewer steps so far.

Results

Steps	Pong reward	Breakout reward	Dead neurons	CKA drift
0	0.0	0.0	47.1%	0.010
500 k	−10.6	4.1	37.1%	0.040
1 M	−7.6	3.5	36.5%	0.047
1.5 M	−7.2	3.2	50.4%	0.045
2 M (final)	−6.8	3.2	62.1%	0.039

Table 2: Interleaved v3 metrics logged every 100 k total steps. CKA measures similarity to the original Pong DQN backbone (higher = more similar).

Interpretation

- **Pong is meaningfully retained.** Final Pong reward of -6.8 sits between the freeze-conv-only baseline (-14.0) and the freeze-all baseline ($+7.2$). The shared backbone learns to serve both tasks without complete forgetting, despite no explicit regularisation or freezing.
- **Breakout learns normally.** Final Breakout reward of 3.2 matches the standalone Breakout DQN baseline (3.86), showing joint training does not impair new game acquisition.
- **Comparison with v2.** The previous step-level design (v2) resulted in Pong collapsing to -21 and Breakout stalling at 0.4 — both tasks failed. The episode-level joint buffer design resolves both failures. This strongly validates the redesign you suggested.
- **Dead neurons increase late in training.** Dead neuron fraction drops from 47% initially to $\sim 37\%$ at 1 M steps (both games partially active), then rises to 62% by 2 M. This late-phase increase may reflect the backbone over-specialising as both games individually approach convergence. The level remains well below the sequential no-freeze condition (87%).
- **CKA drift is small and stable.** The backbone representation stays close to the original Pong reference throughout. Note these networks were trained from scratch, not initialised from the Pong checkpoint, so low CKA drift means both training regimes converge to similar representational geometry.

Experiment 2: Sequential Transfer with Dead Neuron Reinitialisation

Design

Following the sequential training discussion: after training Pong for 2 M steps, identify dead neurons in `fc_repr` and reinitialise them before starting Breakout training.

Dead neuron criterion: A neuron is declared dead if its post-ReLU activation is zero in $>95\%$ of 1 000 sampled Pong frames.

Reinitialisation: The weight rows of dead neurons in `fc_repr` are reset using orthogonal initialisation (gain = $\sqrt{2}$), biases to zero — the same scheme used at network creation. The Adam optimiser is also recreated to clear stale momentum associated with the reinitialised weights.

Result of reinit: 388 out of 512 neurons (75.8%) were dead after Pong training. All 388 were reinitialised.

Budget: Breakout trained for 2 M steps under identical conditions to the no-freeze sequential baseline. Pong retention measured every 200 k steps via chimera evaluation (current backbone + original frozen Pong head).

Results

Breakout steps	Pong reward (chimera)	Breakout reward	Dead neurons	CKA drift
0 (post-reinit, pre-training)	−20.9	0.0	33.0%	0.065
200 k	−21.0	2.1	47.5%	0.018
600 k	−21.0	3.9	57.8%	0.031
1 M	−21.0	3.4	63.1%	0.018
1.6 M	−21.0	5.2	71.9%	0.028
2 M (final)	−21.0	3.5	77.5%	0.025

Table 3: Sequential reinit metrics. Pong reward is the chimera evaluation: current backbone weights combined with the original frozen Pong output head.

Interpretation

- **Reinitialisation did not prevent forgetting.** Pong reward is at floor (−21) immediately after the reinit step (step 0), before any Breakout training has occurred. This is because reinitialising `fc_repr` breaks the backbone’s ability to produce the representations that the Pong head was trained to use, even though the Pong head weights are intact. The forgetting is structural, not a consequence of Breakout overwriting features — it is caused by the reinit itself.
- **Dead neurons grow back rapidly.** After reinit, the dead neuron fraction drops to 33% (from 76% pre-reinit). But by 2 M Breakout steps, it has climbed back to 77.5% — *higher* than where Pong left it. Breakout training drives even more neurons to zero than Pong did. This suggests that dead neuron accumulation is not a pathology specific to Pong, but a general property of deep RL training with ReLU networks.
- **Breakout learns normally.** Breakout reward reaches 3.5 at 2 M steps, comparable to the no-freeze sequential baseline and the standalone DQN, suggesting the reinit does not accelerate or impair Breakout acquisition.
- **Connection to continual backpropagation.** The result illustrates why one-shot reinitialisation at the task boundary is insufficient: it destroys existing useful representations without providing a mechanism for preventing their re-collapse during the new task. Continual backpropagation (Elsayed & Mahmood 2024; Dohare et al. 2024) addresses this by continuously maintaining neuron utility throughout training, rather than intervening once.

Plots

Summary Table

Open Questions for Next Meeting

1. **Terminology:** You referred to lower layers as “representation layers” and higher layers as “strategy/policy layers.” Our codebase uses `fc_repr` as the “representation layer.” Should we standardise to your framing before revising the report?

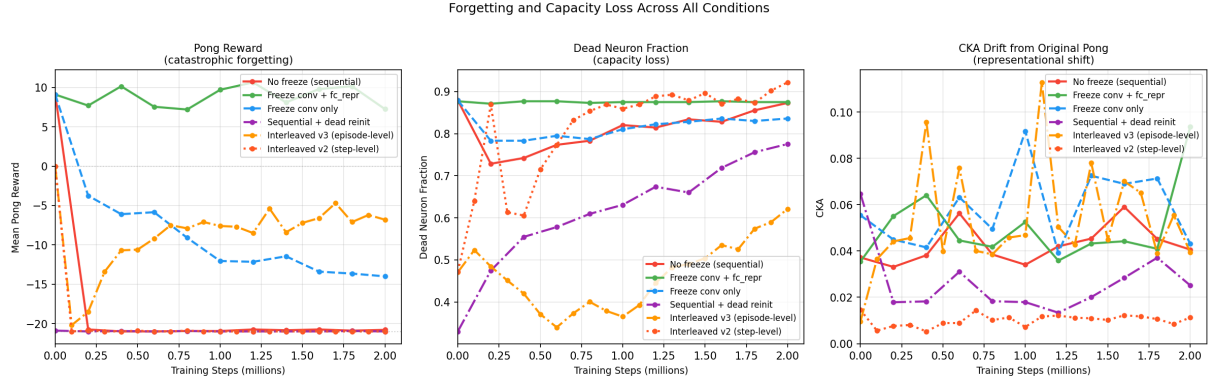


Figure 1: All conditions compared: Pong reward (left), dead neuron fraction (centre), and CKA drift from original Pong backbone (right). Sequential conditions use Breakout training steps on the x-axis; interleaved conditions use total steps.

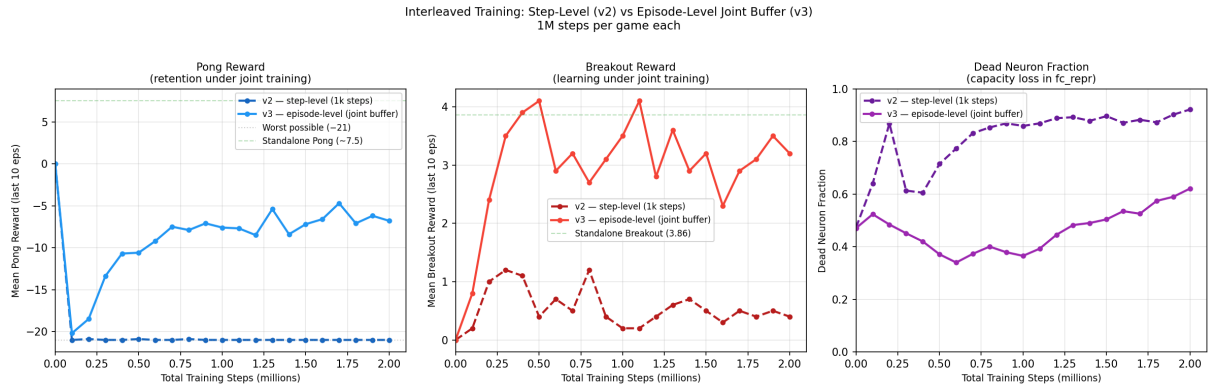


Figure 2: Interleaved v2 (step-level) vs v3 (episode-level, joint buffer). Left: Pong retention. Centre: Breakout learning. Right: dead neuron fraction. v3 succeeds at both tasks; v2 fails at both.

2. **“Predate, make consistent”**: Could you clarify what you meant by this? We interpreted it as updating the report for consistency but are unsure of the exact scope.
3. **CKA probe set**: For cross-game CKA, on-policy frames from one agent are off-distribution for the other. We used random frames as a neutral probe set. Is this justification acceptable, or do you have a preferred approach?
4. **Page 16 / convergence**: Were you referring to the backbone experiment (5 M steps) or the freeze-all sequential condition? Should we re-run to 10 M steps or add a caveat?

Code: <https://github.com/vinoltauro/rl-project>

New experiment scripts: `experiments/interleaved.v3.py`, `experiments/sequential_reinit.py`

Condition	Pong (final)	Breakout (final)	Dead (final)
Sequential no freeze (baseline)	-20.8	3.9	87.3%
Sequential freeze conv only	-14.0	3.7	83.6%
Sequential freeze all	+7.2	—	87.5%
Sequential + dead neuron reinit (new)	-21.0	3.5	77.5%
Interleaved v2 step-level	-21.0	0.4	92.2%
Interleaved v3 episode-level (new)	-6.8	3.2	62.1%
Standalone Pong DQN (reference)	+7.5	—	—
Standalone Breakout DQN (reference)	—	3.86	—

Table 4: Final-checkpoint comparison across all conditions. Pong reward for sequential conditions is chimera evaluation; for interleaved v3 it is online evaluation (no Pong head was transferred). Breakout reward is the online agent mean over last 10 episodes.